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2 **Supplementary Information for**
3 **A reinforcement-based mechanism for discontinuous**
4 **learning**

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Supporting Information Text

1. Maximum entropy inverse reinforcement learning

In this Section, we present a brief introduction to Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL). See refs. (1, 2) and references therein for more details. MaxEnt IRL aims to find a reward function $r(s, a)$ for each state-action pair (s, a) that guides a policy consistent with a set of L observed behavioral trajectories $\mathcal{D} = \{d_1, d_2, \dots, d_L\}$. Each trajectory has $T + 1$ state-action pairs: $d_i = \{s_0^i, a_0^i, s_1^i, \dots, a_{T-1}^i, s_T^i, a_T^i\}$. The transition matrix $P(s'|s, a)$ is assumed to be known.

While the end goal is to estimate $r(s, a)$, MaxEnt IRL proceeds by formulating an unsupervised learning task. Specifically, consider a generative model which assigns a probability $p(\tau)$ for trajectory τ such that the expected discounted reward, $\langle \sum_{t=0}^{T-1} r(s_t, a_t) \gamma^t \rangle_\tau$ from the model matches the empirical discounted reward (note $0 \leq \gamma < 1$ is the standard RL discount factor). To do this, it is sufficient to ensure that the expected frequencies of all s, a pairs appropriately discounted match the empirical ones:

$$\sum_{\tau} p(\tau) \mathbb{1}_{\tau}(s, a) = \frac{1}{L} \sum_{i=1}^L \mathbb{1}_{d_i}(s, a) \quad [1]$$

where the sum over τ is over all possible trajectories with $T + 1$ state-actions pairs, $\mathbb{1}_d(s, a) \equiv \sum_{t=0}^T \delta(s, s_t) \delta(a, a_t) \gamma^t$ for a trajectory $d = (s_0, a_0, s_1, a_1, \dots, s_T, a_T)$ and δ is the indicator function. Since multiple reward functions satisfy Eq. (1), the inverse RL problem is ill-posed without additional constraints. MaxEnt IRL introduces an additional constraint by choosing the generative model $p(\tau)$ which satisfies Eq. (1) and minimizes the relative entropy between $p(\tau)$ and a model $u(\tau)$ whose trajectories are generated by a random policy. In other words, MaxEnt IRL selects the “most random” policy which also satisfies Eq. (1).

$p(\tau)$ is found by minimizing a variational objective with constraints imposed using Lagrange multipliers:

$$\begin{aligned} \mathcal{C} &= \sum_{\tau} p(\tau) \log \frac{p(\tau)}{u(\tau)} - \sum_{s,a} \lambda(s, a) \left(\sum_{\tau} p(\tau) \mathbb{1}_{\tau}(s, a) \right) \\ &\quad - \mu \left(\sum_{\tau} p(\tau) - 1 \right), \\ &= \sum_{\tau} p(\tau) \log \frac{p(\tau)}{u(\tau)} - \sum_{\tau} p(\tau) \left(\sum_{t=0}^T \lambda(s_t, a_t) \gamma^t \right) \\ &\quad - \mu \left(\sum_{\tau} p(\tau) - 1 \right), \end{aligned} \quad [2]$$

where $u(\tau)$ is defined below, the $\lambda(s, a)$'s are Lagrange multipliers which enforce Eq. (1) and μ enforces normalization. Minimizing $\mathcal{C}$ w.r.t $p(\tau)$ gives

$$\begin{aligned} p(\tau) &= \frac{u(\tau) e^{\sum_{t=0}^T \gamma^t \lambda(s_t, a_t)}}{Z}, \quad \text{where} \\ Z &= \sum_{\tau} u(\tau) e^{\sum_{t=0}^T \gamma^t \lambda(s_t, a_t)}, \\ u(\tau) &= p(s_0) \sum_{t=0}^{T-1} P(s_{t+1} | s_t, a_t), \end{aligned} \quad [3]$$

33 and $p(s_0)$ is the probability of the initial state. $u(\tau)$ is (up to a constant factor) the probability of
 34 trajectory τ under a policy which picks actions with equal probability. Observe that the discounted
 35 sum in the exponent of $p(\tau)$ is precisely the discounted long-term reward along τ . The Lagrange
 36 multipliers λ are thus interpreted as the rewards, $r(s, a) \rightarrow \lambda(s, a)$ for each s, a . $p(\tau)$ implicitly
 37 places an exponentially larger weight on policies that lead to rewarding trajectories relative to the
 38 uniform random policy corresponding to u .

The rewards λ are obtained by maximizing the log-likelihood $\mathcal{L}$ of the model over the sample trajectories:

$$\mathcal{L} = \frac{1}{L} \sum_{i=1}^L \log p_{\lambda}(d_i), \quad [4]$$

$$= \frac{1}{L} \sum_{i=1}^L \left(\sum_{t=0}^T \gamma^t \lambda(s_t^i, a_t^i) + \log u(d_i) \right) - \log Z_{\lambda} \quad [5]$$

where the subscript λ is introduced to highlight the dependence on λ . From Eq. (3), we find

$$\frac{\partial \mathcal{L}}{\partial \lambda(s, a)} = \frac{1}{L} \sum_{i=1}^L \mathbb{1}_{d_i}(s, a) - \sum_{t=0}^T \gamma^t p_{\lambda,t}(s, a), \quad [6]$$

39 where $p_{\lambda,t}(s, a)$ is the marginal probability of encountering state-action pair s, a at time t w.r.t p .
 40 The Markovianity of the process enables efficient calculation of $p_{\lambda,t}(s, a)$ using a forward-backward
 41 algorithm (described below) similar to the one used to train Hidden Markov Models (HMM). When
 42 $\mathcal{L}$ attains its optimum, we observe from Eq. (6) that the constraint Eq. (1) is automatically satisfied.

The forward-backward equations are similar to the HMM case except for an exponential weight:

$$\begin{aligned} \alpha_k(s, a) &= \sum_{s', a'} P(s|s', a') e^{\gamma^k \lambda(s, a)} \alpha_{k-1}(s', a'), \\ \beta_k(s, a) &= \sum_{s', a'} P(s'|s, a) e^{\gamma^{T-k} \lambda(s, a)} \beta_{k-1}(s', a') \end{aligned} \quad [7]$$

for $k > 0$ and $\alpha_0(s, a) = p_0(s) e^{\lambda(s, a)}$, $\beta_0(s, a) = e^{\gamma^T \lambda(s, a)}$. We have

$$p_{\lambda,t}(s, a) = \frac{\alpha_t(s, a) \beta_{T-t-1}(s, a)}{Z_{\lambda}}, \quad Z_{\lambda} = \sum_{s, a} \alpha_T(s, a). \quad [8]$$

43 In practice, we compute the logarithm of the various quantities and use $\log \sum_x e^x = \max(x) +$
 44 $\log \left(1 + \sum_{x \neq \max(x)} e^{x - \max(x)} \right)$ to prevent underflow/overflow. The λ s are obtained using gradient
 45 ascent on $\mathcal{L}$ using standard optimization methods.

46 2. The dynamics of sequence learning on tree-structured relational graphs

47 In this Section, we expand on the dynamics of learning in the goal-oriented navigation task intro-
 48 duced in the main text. The dynamics are analyzed in full detail for the SARSA learning algorithm
 49 used in the main text and other commonly used RL rules. These alternative rules also display
 50 wave-like propagation of the value signal, with differences in the speed and front profiles for cer-
 51 tain variants. Part of this Section is reproduced in the Methods, and is included here for completeness.

52

Learning framework. An RL algorithm is defined by the learning rule and the (possibly non-stationary and stochastic) policy executed during learning. We consider a stochastic policy, where for each state-action pair (s, a) , we have $\ln \pi(a|s) = q(s, a) - \ln Z(s)$, where $q(s, a) = q_\varepsilon(s, a) + q_r(s, a)$ and $Z(s)$ is a normalization constant. $q_\varepsilon(s, a)$ represents fixed intrinsic rewards that drive exploration and determine the agent's exploratory biases. The extrinsic rewards $q_r(s, a)$ (initially set to zero) are modulated by the learning rule. At the beginning of learning, the agent's behavior depends solely on its exploratory biases. As learning proceeds and the agent acquires rewards, its behavior is modified by the increasing effect of the extrinsic rewards. We consider the learning dynamics for various learning rules discussed further below.

Setup and notation. The linear track consists of $N-1$ nodes on the direct path, $n = 1, 2, \dots, N-1$. The agent starts each episode at node $n = N$ and the reward is at the goal node $n = 0$. In addition to these nodes, the nodes from $n = 1$ to $N-1$ each have a side path, which we label as $1_b, 2_b, \dots, (N-1)_b$. The state space of the Markov decision process is the set of *directed edges* that connect the various nodes and the side paths as shown in Figure 3a. In other words, both the agent's location in the graph and the direction in which it is headed matter. We denote (n_1, n_2) as the directed edge from n_1 to n_2 .

The transition dynamics $P(s'|s, a)$ are deterministic (note however that the policy $\pi(a|s)$ is stochastic). At each directed edge, the agent can choose to go along the directed edges emanating from its current node, except for turning back, for e.g., the transition $(n+1, n) \rightarrow (n, n+1)$ is disallowed. This simplifying assumption does not affect the results as the agent can effectively turn back by going into a side path and returning $(n+1, n) \rightarrow (n, n_b) \rightarrow (n_b, n) \rightarrow (n, n+1)$. The episode begins with the agent at the directed edge $(N, N-1)$. The directed edge pointing towards the goal node, $(1, 0)$, is an absorbing state, i.e., the agent receives a reward r and the episode ends once the agent traverses that edge. We impose reflecting conditions at edges going into the side paths (n, n_b) and the start node $(N-1, N)$.

The agent receives identical intrinsic exploration rewards at every intersection on the direct path. There are three directed edges leading to any node n , and we thus consider three cases at each node. These three cases are shown pictorially in Figure 3a. Since the agent can take two actions at each step and the policy only depends on differences of q values, we specify the q values for only one of the actions. Note that q values for both actions are taken into account in the full-scale RL simulations. For the purposes of the theoretical analysis discussed below, it suffices to track the q value for only one of the actions as the q value for the other action is rarely reinforced. The notation used for the three cases is introduced (see also Figure 3a).

1. the agent is on the direct path and going towards the goal, $(n+1, n)$: for the action corresponding to the agent continuing towards the goal $(n+1, n) \rightarrow (n, n-1)$, we denote $q_\varepsilon \equiv \varepsilon, q_r \equiv q_n$.
2. the agent is on the side path n_b and going towards n , (n_b, n) : for the action corresponding to the agent turning towards the goal $(n_b, n) \rightarrow (n, n-1)$, we denote $q_\varepsilon \equiv \varepsilon', q_r \equiv q'_n$.
3. the agent is on the direct path and going towards the start, $(n-1, n)$: for the action corresponding to the agent continuing towards the start $(n-1, n) \rightarrow (n, n+1)$, we denote $q_\varepsilon \equiv \varepsilon'', q_r \equiv q''_n$.

The probabilities of taking the action described in each of three cases is denoted $\sigma_n \equiv \sigma(\varepsilon + q_n), \sigma'_n \equiv \sigma(\varepsilon' + q'_n), \sigma''_n \equiv \sigma(\varepsilon'' + q''_n)$, where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic function.

97 **Dynamics of exploration.** We first consider the dynamics of exploration with no learning. An
 98 important byproduct of the calculation is the identification of the values of $\varepsilon, \varepsilon', \varepsilon''$ for which the
 99 agent finds the goal within a reasonable amount of time. These values delineate the parameter
 100 regime where learning occurs (the marching and expanding regimes in Figure 4f) *vs* the regime in
 101 which no learning occurs as the agent rarely finds the goal in the asymptotic limit $N \gg 1$ (the
 102 stalled regime in Figure 4f). The upshot is that in these two cases the agent on average either drifts
 103 linearly towards the goal or constantly reverts back to the start. The marginal case corresponds to
 104 diffusive behavior.

To identify these regimes, we calculate the expected time for the agent, starting at $(N - 1, N)$, to find the goal state. We additionally calculate the expected number of times the agent visits each state which is used further below for the analysis of learning dynamics. The calculation of both of these quantities is simplified by considering the dynamics of a simpler, equivalent Markov chain. Specifically, we observe that when the agent enters a node n , say along $(n + 1, n)$, regardless of whether it goes into the side path or not, it either continues along its path, $(n, n - 1)$, or turns back $(n, n + 1)$. The probability of the former possibility, k_+ , is the sum of the probability that it moves to $(n, n - 1)$ along the direct path, $\sigma(\varepsilon)$, and the probability that it takes a detour through the side path, $(1 - \sigma(\varepsilon))\sigma(\varepsilon')$. The probability of turning back is $1 - k_+$. Similarly, the probability of continuing towards the start state when the agent is at $(n - 1, n)$ is $k_- = \sigma(\varepsilon'') + (1 - \sigma(\varepsilon''))(1 - \sigma(\varepsilon'))$, and the probability of turning back to $(n, n - 1)$ is $1 - k_-$. In summary, the probabilities that the agent continues towards the goal and towards the start are

$$\begin{aligned} k_+ &= \sigma(\varepsilon) + (1 - \sigma(\varepsilon))\sigma(\varepsilon'), \\ k_- &= \sigma(\varepsilon'') + (1 - \sigma(\varepsilon''))(1 - \sigma(\varepsilon')) \end{aligned} \quad [9]$$

105 respectively. The dynamics of the Markov chain defined by these two parameters are easily computed.
 106

(a) *Expected time to the goal.* Let the expected time to the goal starting from $(n, n - 1)$ and $(n - 1, n)$ be denoted $T_+(n)$ and $T_-(n)$ respectively. We aim to compute $T_+(N)$ with an absorbing boundary at $(1, 0)$, so that $T_+(1) = 0$. The reflecting boundary conditions at $(N - 1, N)$ imply $T_-(N) = T_+(N) + 1$. We have

$$T_+(n) = 1 + k_+T_+(n - 1) + (1 - k_+)T_-(n), \quad [10]$$

$$T_-(n) = 1 + k_-T_-(n + 1) + (1 - k_-)T_+(n), \quad [11]$$

for $1 < n \leq N$ and $1 < n < N$ respectively. Expressing $T_-(n)$ and $T_-(n + 1)$ in terms of $T_+(n - 1), T_+(n), T_+(n + 1)$ using Eq. (10) and plugging into Eq. (11), we obtain the second-order difference equation

$$\begin{aligned} k_+(T_+(n) - T_+(n - 1)) - k_-(T_+(n + 1) - T_+(n)) \\ = 2 - k_+ - k_-, \end{aligned} \quad [12]$$

107 for $1 < n < N$. Defining $U(n) = T_+(n + 1) - T_+(n)$ gives

$$108 \quad U(n - 1) = \frac{2 - k_+ - k_-}{k_+} + \frac{k_-}{k_+}U(n). \quad [13]$$

Since $T_-(N) = T_+(N) + 1$, setting $n = N$ in Eq. (10) gives $U(N - 1) = -1 + 2/k_+$. Multiplying

both sides by $(k_-/k_+)^{n-1}$ and summing from $n + 1$ to $N - 1$, we get

$$\begin{aligned} U(n) & \left(\frac{k_-}{k_+} \right)^n \\ &= \frac{2 - k_+ - k_-}{k_+} \sum_{n'=n}^{N-2} \left(\frac{k_-}{k_+} \right)^{n'} + \frac{2 - k_+}{k_+} \left(\frac{k_-}{k_+} \right)^{N-1}, \end{aligned} \quad [14]$$

which after simplification leads to

$$U(n) = \frac{2 - k_+ - k_-}{k_+ - k_-} + \frac{2(1 - k_+)}{k_- - k_+} \left(\frac{k_-}{k_+} \right)^{N-n}, \quad [15]$$

for $k_- \neq k_+$. Summing $U(n)$ from $n = 0$ to $N - 1$ and using $T_+(0) = 0$, we obtain the mean time to find the goal

$$\begin{aligned} T_+(N) &= \left(\frac{2 - k_+ - k_-}{k_+ - k_-} \right) N \\ &+ \frac{2k_-(1 - k_+)}{(k_- - k_+)^2} \left(\left(\frac{k_-}{k_+} \right)^N - 1 \right). \end{aligned} \quad [16]$$

If $k_- > k_+$, the agent takes time exponential in N , which is infeasible for large N . For $k_- < k_+$, the agent travels linearly towards the goal with drift $J = \frac{2-k_+-k_-}{k_+-k_-}$. The marginal case of $k_- = k_+$ leads to diffusive behavior. In this case, from Eq. (14), we have

$$U(n) = \frac{2(1 - k_+)}{k_+} (N - 1 - n) + \frac{2 - k_+}{k_+} \quad [17]$$

Summing from $n = 0$ to $N - 1$ gives $T_+(N) \simeq \frac{1-k_+}{k_+} N^2$.

(b) *Expected number of visits.* We calculate the expected number of times per episode the agent visits each state on the direct path. Denote the expected number of times the agent visits states $(n + 1, n)$ and $(n, n + 1)$ as $M_+(n)$ and $M_-(n)$ respectively. We have

$$M_+(n) = (1 - k_-)M_-(n) + k_+M_+(n + 1), \quad [18]$$

$$M_-(n) = k_-M_-(n - 1) + (1 - k_+)M_+(n), \quad [19]$$

We observe that setting $n \rightarrow n + 1$ in Eq. (19) and adding Eq. (18) gives a conservation equation, $M_+(n) + M_-(n + 1) = M_-(n) + M_+(n + 1)$. At the boundaries we have $M_-(1) = (1 - k_+)M_+(1)$ and $M_+(N - 1) = M_-(N - 1) + 1$. Combining the latter boundary condition and the conservation equation, we see that $M_+(n) = M_-(n) + 1$ for all n . Plugging this into the former boundary condition then leads to $M_+(1) = 1/k_+$ and into Eq. (18) leads to

$$M_+(n + 1) = \frac{1 - k_-}{k_+} + \frac{k_-}{k_+} M_+(n). \quad [20]$$

Dividing both sides by $(k_-/k_+)^{n+1}$, summing from 1 to n and simplifying gives

$$M_+(n) = \frac{1 - k_-}{k_+ - k_-} - \left(\frac{k_-}{k_+} \right)^n \frac{1 - k_+}{k_+ - k_-} \quad [21]$$

for $k_- \neq k_+$. When $k_- = k_+$, we have $M_+(n) = 1 + n(1 - k_+)/k_+$.

Dynamics of learning. We now analyze the dynamics of reinforcement propagation for five learning rules, beginning with the SARSA rule considered in the main text. Throughout, we consider the slow-learning limit $\alpha \ll 1$. In this limit, we can compute the learning dynamics averaged over the agent's behavior in each episode. Simulations show that the analytical results are accurate up to $\alpha \lesssim 0.1$, with wave-like propagation of the reinforcement signal observed for even larger α . In all cases analyzed below, we set the standard discount factor in RL (γ) to unity. This discount factor introduces an effective length scale for the influence of the reward and regularizes the Bellman equation, which are not necessary in our setting due to the regularization and effective horizon provided by the exploration-based stochastic policy. In our analysis below, it is easy to see that the effective horizon is $N_{\text{range}} \simeq (-\ln \sigma(\varepsilon + r))^{-1}$. For $N \ll N_{\text{range}}$, the reinforcement signal propagates as a wave. The wave slows down and the reinforcement signal eventually decays to zero at steady state when $N \gg N_{\text{range}}$. We consider the $1 \ll N \ll N_{\text{range}}$ limit in our analysis below, which thus requires that $\varepsilon + r$ be appropriately large. In the linear track setting, reward is acquired only when the agent reaches the goal state. This is set by the boundary condition $q_0 = r$ (see the setup and notation section above). The initial conditions are $q_n = q'_n = q''_n = 0$ for $n > 0$. Also note that q''_n , i.e., the q_r value corresponding to going *away* from the goal is never reinforced and stays fixed at $q''_n = 0$. We thus consider the dynamics of q_n and q'_n in what follows.

(a) **SARSA.** Whenever the agent makes a state-action-state transition s_t, a_t, s_{t+1} , the SARSA(0) (state-action-reward-state-action) learning rule updates the corresponding q_r values as

$$\begin{aligned} q_r(s_t, a_t) &\rightarrow q_r(s_t, a_t) + \alpha \delta_t, \quad \text{where} \\ \delta_t &= r_t + \langle q_r(s_{t+1}, \cdot) \rangle_{\pi(\cdot|s_{t+1})} - q_r(s_t, a_t), \text{ if } s_{t+1} \text{ is not the goal} \\ \delta_t &= r_t - q_r(s_t, a_t), \text{ if } s_{t+1} \text{ is the goal.} \end{aligned} \tag{22}$$

Here $0 < \alpha < 1$ is the learning rate, r_t is the reward obtained at time t and the expectation is over future actions drawn from the policy π . As discussed previously, $\ln \pi(a|s) = q_\varepsilon(s, a) + q_r(s, a) - \ln Z(s)$. Note that the learning rule described above is strictly speaking the ‘‘expected SARSA’’ rule (3), which is an efficient variant of the SARSA(0) rule. The 0 in the parenthesis corresponds to the eligibility trace parameter $\lambda = 0$, which converts the local SARSA(0) rule to a non-local rule for $\lambda > 0$. We discuss eligibility traces further below.

Applying the SARSA rule Eq. (22), the expected change Δq_n and $\Delta q'_n$ in each episode is given by

$$\begin{aligned} \Delta q_n &= \alpha \mu_n (\sigma_{n-1} q_{n-1} - q_n), \\ \Delta q'_n &= \alpha \mu'_n (\sigma_{n-1} q_{n-1} - q'_n), \end{aligned} \tag{23}$$

for $n > 1$ and $\Delta q_1 = \alpha \mu_1 (r - q_1)$, $\Delta q'_1 = \alpha \mu'_1 (r - q'_1)$. The $\sigma_{n-1} q_{n-1}$ term is the expected value of q_r at the subsequent state, i.e., once the agent crosses node n . Here μ_n and μ'_n are the expected number of times per episode the agent crosses node n towards the goal along the direct path or by making a detour through the side path at n , respectively. The relative probability of these two events determines their ratio, $\mu_n/\mu'_n = \sigma_n/(1 - \sigma_n)\sigma'_n$, and the total expected crossings through node n towards the goal determines their sum $\mu_n + \mu'_n$. The expected crossings at each node depends in general on the q_ε and q_r values at all states. It is possible to calculate the expected crossings recursively similar to the case of pure exploration above. However, we note that for the nodes where learning occurs (near the front of the wave), we should expect a single crossing, i.e., $\mu_n + \mu'_n \simeq 1$.

As argued in the main text, the key idea is that learning at node n only occurs when $\sigma_{n-1}q_{n-1}$ is non-negligible, i.e., the probability of staying on the direct path is sufficiently reinforced at $n-1$ and thus also for $n' \leq n-1$. Since these actions are reinforced, the agent is very likely to take the direct path to the goal immediately after crossing n . Equivalently, whenever the agent crosses n towards the goal, the probability of it cycling back to $n' > n$ and making another attempt at crossing n is small, leading to $\mu_n + \mu'_n \simeq 1$. We thus obtain

$$\mu_n = \frac{\sigma_n}{\sigma_n + (1 - \sigma_n)\sigma'_n}, \quad \mu'_n = 1 - \mu_n \quad [24]$$

We can convert the discrete dynamics equations above to a continuous-time equation in the limit $\alpha \ll 1$, to get

$$\begin{aligned} \dot{q}_n &= \mu_n (\sigma_{n-1}q_{n-1} - q_n), \\ \dot{q}'_n &= \mu'_n (\sigma_{n-1}q_{n-1} - q'_n), \end{aligned} \quad [25]$$

where the dot represents a time derivative and the unit of time is $1/\alpha$. Integrating the above equations over a unit time interval $\Delta t = 1$ corresponds to expected changes in q_n and q'_n over $1/\alpha$ episodes.

As shown in Figure 4a, integrating the equations Eq. (25) provides an accurate approximation of the values obtained through full-scale RL simulations. We expand on the analysis of Eq. (25) presented in the main text. Below, we consider exploration parameter values, $\varepsilon, \varepsilon', \varepsilon''$, such that $k_- < k_+$, i.e., in the regime where the agent consistently gets to the goal and learning occurs. We set $\varepsilon'' = 0$ and vary $\varepsilon, \varepsilon'$. The key parameter which determines the character of the wave is ε . ε' is chosen such that $k_- < k_+$. It is useful to consider the asymptotic limits $e^\varepsilon \gg 1$ (expanding regime) and $e^{-\varepsilon} \gg 1$ (marching regime). The motivation for the names will become apparent.

Expanding regime: The expanding regime has straightforward linear learning dynamics. To see this, note that when $e^\varepsilon \gg 1$, we have $\sigma_n \simeq 1$ for all n . This implies from Eq. (24) that $\mu_n \simeq 1$, $\mu'_n = 1 - \mu_n$ is negligible and q'_n thus remains at 0. From Eq. (25)

$$\dot{q}_n = q_{n-1} - q_n, \quad [26]$$

with boundary conditions $q_0(t) = r$ and initial conditions $q_n(0) = 0$ for $n > 0$. Defining $\tilde{q}_n(s) = \int_0^\infty q_n(t)e^{-st}dt$, i.e., the Laplace transform of $q_n(t)$, Eq. (26) leads to

$$(s+1)\tilde{q}_n(s) = \tilde{q}_{n-1}(s) + q_n(0). \quad [27]$$

Multiplying both sides of the equation by $(s+1)^{n-1}$ and summing the recursion gives

$$(s+1)^n \tilde{q}_n(s) = \tilde{q}_0(s) + \sum_{m=1}^n q_m(0)(s+1)^{m-1}. \quad [28]$$

Using $\tilde{q}_0(s) = r/s$ and performing the inverse Laplace transform leads to

$$q_n(t) = rP(n, t) + \sum_{m=1}^n \frac{t^{n-m}e^{-t}}{(n-m)!} q_m(0), \quad [29]$$

where $P(n, t)$ is the regularized lower incomplete gamma function. Setting the boundary conditions, we have $q_n(t) = rP(n, t)$. For large n it is well known that the gamma distribution is approximated

by a normal distribution: $P(n, t) \approx \Phi\left(\frac{t-n}{\sqrt{n}}\right)$, where Φ is the standard normal cdf. The half-maximum of the propagating signal thus travels as $n_{1/2} = t$, i.e., with speed $v_0 = 1$ corresponding to 1 node every $1/\alpha$ episodes. The width of the front expands as $\sqrt{n_{1/2}} = \sqrt{t}$.

Marching regime: Qualitatively different behavior is observed in the opposite asymptotic limit, $e^{-\varepsilon} \gg 1$. In this case, the nonlinear dynamics of Eq. (25) come into play. We show below that the dynamics consist of a self-similar wavefront which marches forward one node at a time, with a fixed time interval, τ_0 , between each node (and thus its speed is $v_0 = \tau_0^{-1}$). We first present the qualitative picture and then proceed to calculate τ_0 . It turns out that a complete characterization of the wave dynamics is feasible, which leads to analytical expressions for τ_0 , the shape of the wavefront and the dynamics of reinforcement at each node, $q_n(t)$.

Since $e^{-\varepsilon} \gg 1$, the probability that the agent takes the direct path through node n , μ_n , is negligible before any reinforcement occurs at node n . Any reinforcement that occurs is due to rare events which gradually increase q_n provided that $q_{n-1} > |\varepsilon|$. This bottleneck remains until q_n itself gradually increases to $q_n \simeq |\varepsilon|$. Meanwhile, all the actions on the direct path for $n' < n$ are continuously reinforced. This reinforcement is possible as the agent can bypass crossing n through the direct path by instead taking a detour through the side path at n . Indeed, learning occurs ($k_- < k_+$) in the marching regime only if ε' is sufficiently large to allow for frequent detours through the side path. As a consequence, while q_n is gradually being reinforced, q'_n is sufficiently reinforced so that $\sigma'_n \simeq 1$ and $\mu_n \simeq \sigma_n$ (from Eq. (24)). The detours through the side paths ensure that q_{n-1} is rapidly reinforced and $r \gg |\varepsilon|$ implies that σ_{n-1} displays switch-like behavior once $q_{n-1} > |\varepsilon|$. Since q_{n-1} is increasing, successive rare events through the direct path at node n receive increasing amounts of reinforcement on q_n . Calculating the amount of reinforcement on q_n thus requires knowing the dynamics of q_{n-1} . However, the dynamics of q_{n-1} depend on the dynamics of q_{n-2} , which in turn depends on q_{n-3} and so on, leading to cascading dependencies. The duration τ_0 for q_n to be reinforced from 0 to $|\varepsilon|$ will thus depend on interactions with the bulk of the wave. We now calculate τ_0 by solving the equation hierarchy. We will see that the solution is feasible due to the linear dynamics in the bulk of the wave Eq. (26), which considerably simplify the analysis.

We begin ($t = 0$) from the moment when the wave has just reached node n , i.e., $q_{n-1}(0) = |\varepsilon|$ and $q_n(0) = 0$ begins to be reinforced. τ_0 is the time it takes for one cycle to complete, which is when $q_n(\tau_0) = |\varepsilon|$ is reached and q_n begins to influence q_{n+1} . From Eq. (25), the (approximate) evolution of q_n during this interval is given by

$$\dot{q}_n = \sigma_n q_{n-1}, \quad [30]$$

where we have used $\mu_n \simeq \sigma_n$ and $\sigma_{n-1} \simeq 1$. Since $r \gg |\varepsilon|$ implies $q_n \ll q_{n-1}$ in this period, we have also ignored the negative feedback due to q_n . This argument can be made rigorous by computing q_n from Eq. (30) (see below), which will lead to an upper bound on q_n , and showing that this upper bound is $\ll q_{n-1}$. Integrating Eq. (30) for $t \leq \tau_0$, we get

$$\int_0^{q_n(t)} \frac{dq_n}{\sigma_n} = \int_0^t q_{n-1}(t') dt' = x_t, \quad [31]$$

where $x_t \equiv \int_0^t q_{n-1}(t') dt'$ is to be calculated. Plugging in $1/\sigma_n = 1 + e^{|\varepsilon| - q_n}$ and integrating gives

$$q_n(t) + e^{|\varepsilon|} (1 - e^{-q_n(t)}) = x_t. \quad [32]$$

The solution of the above equation is expressed in terms of the Lambert W function (4), $W(x)$,

$$q_n(t) = W(e^{|\varepsilon| + e^{|\varepsilon|} - x_t}) + x_t - e^{|\varepsilon|}. \quad [33]$$

230 Since $q_n(\tau_0) = |\varepsilon|$, τ_0 is obtained from solving

$$231 \quad |\varepsilon| + e^{|\varepsilon|} - 1 = x_{\tau_0}. \quad [34]$$

232 We now calculate x_t . In the bulk ($m < n$), we have

$$233 \quad \dot{q}_m = q_{m-1} - q_m. \quad [35]$$

234 From Eq. (29),

$$235 \quad q_m(t) = rP(m, t) + \sum_{m'=1}^m \frac{t^{m-m'} e^{-t}}{(m-m')!} q_{m'}(0). \quad [36]$$

236 Note that the initial conditions $q_{m'}(0)$ here are the unknown values of $q_{m'}$ when $q_{n-1} = |\varepsilon|$ and
237 $q_n = 0$, which are to be computed self-consistently under the marching dynamics with time step τ_0 .

238 Using the series expansion of $P(m, t)$ and $m' \rightarrow m - m'$, we re-write the above equation as

$$239 \quad (r - q_m(t)) = \sum_{m'=0}^{m-1} \frac{t^{m'} e^{-t}}{m'!} (r - q_{m-m'}(0)). \quad [37]$$

240 Here increasing values of m' correspond to the nodes reinforced in earlier stages of the learning
241 process. Thus, the terms $r - q_{m-m'}(0)$ are decreasing in magnitude and the terms of large m' do not
242 matter for $\tau_0 > 1$. We may then take the sum to infinity in the equation above.

243 To solve for $q_m(0)$ for $m < n$, we notice that the self-similarity of the wavefront with period τ_0
244 implies $q_m(\tau_0) = q_{m-1}(0)$. Plugging $t = \tau_0$ into Eq. (37), using $q_m(\tau_0) = q_{m-1}(0)$, we get

$$245 \quad g(m-1) = \sum_{m'=0}^{\infty} \frac{\tau_0^{m'} e^{-\tau_0}}{m'!} g(m-m'), \quad [38]$$

246 where $g(m) \equiv 1 - q_m(0)/r$. This difference equation applies for all $m < n$. Since the difference
247 equation has constant coefficients, solutions are of the form $g(m) = cb^m$, where c is a constant (5).

248 We have

$$249 \quad b^{m-1} = \sum_{m'=0}^{\infty} \frac{\tau_0^{m'} e^{-\tau_0}}{m'!} b^{m-m'} = b^m e^{-\tau_0} e^{\tau_0/b}. \quad [39]$$

250 Thus b satisfies

$$251 \quad be^{\tau_0/b} = e^{\tau_0} \quad [40]$$

252 which leads to

$$253 \quad b = \frac{-\tau_0}{W(-\tau_0 e^{-\tau_0})}, \quad [41]$$

where W is the non-trivial branch of the Lambert W function (the trivial branch evaluates to 1).
Evaluating Eq. (37) for $m = n-1$ then yields

$$\begin{aligned} q_{n-1}(t) &= r \left(1 - c \sum_{m'=0}^{n-1} \frac{t^{m'} e^{-t}}{m'!} b^{n-1-m'} \right) \\ &= r \left(1 - cb^{n-1} e^{-t+t/b} \right). \end{aligned} \quad [42]$$

254 Since $q_{n-1}(0) = |\varepsilon|$, we get $cb^{n-1} = 1 - |\varepsilon|/r$. The evolution of $q_{n-1}(t)$ is thus

$$255 \quad q_{n-1}(t) = r - (r - |\varepsilon|)e^{-\beta t}, \quad [43]$$

where $\beta \equiv 1 - 1/b = 1 + \tau_0^{-1}W(-\tau_0 e^{-\tau_0})$, which determines the wave profile at any instant t since the periodicity implies $q_{n-n'}(t) = q_{n-1}(t + (n' - 1)\tau_0)$.

Note that $W(x)$ is real-valued for $x \geq -1/e$, which corresponds to $\tau_0 \geq 1$. For $\tau_0 = 1$, using $W(-1/e) = -1$ we see that $\beta = 1$ and the power solution $g(m) = c\beta^m$ cannot be used. It can be checked that $g(m) = cm$ satisfies the recursion (c is again a constant). The initial conditions yield $c = \frac{1-|\varepsilon|/r}{n-1}$, whose dependence on n violates the self-similarity of the wavefront. Thus, the dynamics when $\tau_0 = 1$ cannot be described by a self-similar traveling wave. Indeed, as shown in the expanding regime section, when $\tau_0 = 1$ the width of the wavefront expands as $\sqrt{t}$.

Using the identity $e^{W(x)} = x/W(x)$, we obtain $x_{\tau_0} = \int_0^{\tau_0} q_{n-1}(t)dt = r(\tau_0 - 1) + |\varepsilon|$. Plugging this into Eq. (34), we finally have

$$\tau_0 = \frac{r + e^{|\varepsilon|} - 1}{r}. \quad [44]$$

An interesting alternative derivation of τ_0 exploits a conservation equation for x_{τ_0} . Since $\dot{q}_{n-1} = q_{n-2} - q_{n-1}$, integrating both sides from 0 to τ_0 and re-arranging yields

$$\begin{aligned} \int_0^{\tau_0} q_{n-1}dt &= \int_0^{\tau_0} q_{n-2}dt - (q_{n-1}(\tau_0) - q_{n-1}(0)) \\ &= \int_0^{\tau_0} q_{n-3}dt - (q_{n-2}(\tau_0) - q_{n-2}(0)) \\ &\quad - (q_{n-1}(\tau_0) - q_{n-1}(0)) \\ &= \int_0^{\tau_0} q_{n-3}dt - (q_{n-2}(\tau_0) - q_{n-1}(0)). \end{aligned} \quad [45]$$

where we have used the periodicity in the bulk, $q_{n-2}(0) = q_{n-1}(\tau_0)$ in the last step. Repeating this sequence of steps, we are lead to

$$\int_0^{\tau_0} q_{n-1}dt = \int_0^{\tau_0} q_0dt - (q_0(\tau_0) - q_{n-1}(0)) = r(\tau_0 - 1) + |\varepsilon|. \quad [46]$$

(b) **Eligibility traces.** In RL, eligibility traces are used to enhance learning speed by efficiently propagating errors backwards in time. Specifically, the method uses the reward prediction error (δ_t in Eq. (22)) at the current state-action pair to update the q_r values of recently visited state-action pairs, in addition to the q_r values of the current state-action pair. The resultant q_r update of the state-action pair visited j steps before the current one is

$$q_r(s_{t-j}, a_{t-j}) \rightarrow q_r(s_{t-j}, a_{t-j}) + \alpha(\gamma\lambda)^j \delta_t, \quad [47]$$

where $0 \leq \lambda < 1$ is the eligibility traces parameter and γ is the discount factor (recall that γ is set to 1 throughout our analysis). λ sets the effective number of previous state-action pairs which are affected by the update at the current state-action pair. We consider a slightly modified version of this learning rule, where λ is set to 1 and instead a fixed number k of previous state-action pairs are updated according to Eq. (47).

Let us examine the effect of this rule when updating q_n (the case of q'_n follows). If, after crossing node n towards the goal, the agent goes into the side path at $n - 1$, then since all the “hidden” state-action pairs inside the side path have $q_r = 0$ and the side paths are assumed to be long detours ($\gg k$), the updates within the side path and after exiting the side path do not update q_n . In this

case, q_n is updated just as in Eq. (25). If the agent, instead of turning into the side path at $n - 1$, continues along the direct path towards node $n - 2$, and if $k \geq 1$, then q_n is updated as

$$\Delta q_n = \alpha ((\sigma_{n-1} q_{n-1} - q_n) + (\sigma_{n-2} q_{n-2} - q_{n-1})), \quad [48]$$

where the first term in the parenthesis is the δ -error from when the agent crosses node n and the second term is from when the agent crosses $n - 1$ along the direct path. If, at this point, the agent turns into the side path at $n - 2$, there are no further updates of q_n for the same reason stated above. Similarly, if it instead continues along the direct path to $n - 3$ and $k \geq 2$, then q_n is again updated using the δ -error at that transition. Extending this argument, we see that q_n (and q'_n by the same argument) receives non-local updates as long as the agent continues along the direct path without taking turns into side paths. In the limit of $\alpha \ll 1$, we take expectations to get

$$\begin{aligned} \dot{q}_n &= \mu_n \left((\sigma_{n-1} q_{n-1} - q_n) + \sum_{k'=1}^k \left(\prod_{k''=1}^{k'} \sigma_{n-k''} \right) (\sigma_{n-k'-1} q_{n-k'-1} - q_{n-k'}) \right), \\ &= \mu_n \left(\left(\prod_{k'=1}^{k+1} \sigma_{n-k'} \right) q_{n-k-1} - q_n \right), \\ \dot{q}'_n &= \mu'_n \left(\left(\prod_{k'=1}^{k+1} \sigma_{n-k'} \right) q_{n-k-1} - q'_n \right) \end{aligned} \quad [49]$$

The product, $\prod_{k'=1}^k \sigma_{n-k'}$, corresponds to the probability of taking k contiguous steps along the direct path. The second equation is obtained by noticing that all the terms except the ones involving q_n and q_{n-k-1} cancel out. We verify the validity of Eq. (49) by comparing to full RL simulations (Figure S4a).

As in the SARSA case, we consider the asymptotic limits $e^\varepsilon \gg 1$ and $e^{-\varepsilon} \gg 1$. For $e^\varepsilon \gg 1$, we have $\sigma_n \simeq 1, \mu_n \simeq 1$ for all n and the sums in Eq. (49) collapse into the simple set of linear equations

$$\dot{q}_n = q_{n-(1+k)} - q_n, \quad \dot{q}'_n = 0. \quad [50]$$

For convenience, we assume $n = \ell(1 + k)$ where ℓ is an integer. Similar results with minor modifications are obtained for the other cases. Taking the Laplace transform, we get

$$(s + 1) \tilde{q}_{\ell(1+k)} = \tilde{q}_{(\ell-1)(1+k)}(s) + q_{\ell(1+k)}(0), \quad [51]$$

which after following steps similar to the SARSA case leads to

$$q_{\ell(1+k)}(t) = rP(\ell, t) + \sum_{\ell'=1}^{\ell} \frac{t^{\ell-\ell'} e^{-t}}{(\ell - \ell')!} q_{\ell'(1+k)}(0). \quad [52]$$

For the initial conditions $q_n(0) = 0$ for $n > 0$, we have $q_n(t) = rP(\ell, t) = rP\left(\frac{n}{1+k}, t\right) \approx \Phi\left(\frac{(1+k)t-n}{\sqrt{(1+k)n}}\right)$ for large n . From here, we obtain the speed, $v_k = 1 + k$, and the width of the wavefront at time t , $(1 + k)\sqrt{t}$. Thus, eligibility traces enhance the speed of reinforcement propagation by breaking the locality constraint of SARSA.

In the marching regime, $e^{-\varepsilon} \gg 1$, suppose, as in the SARSA case, that the wave has just reached node n , i.e., $q_{n-1}(0) = |\varepsilon|$, $q_n(0) = 0$. We compute the time τ_k it takes for q_n to be reinforced to $|\varepsilon|$. When $q_{n-1} \geq |\varepsilon|$, from Eq. (41), the product of σ 's equals 1 and $q_n(t)$ subsequently evolves as

$$\dot{q}_n \simeq \sigma_n q_{n-k-1}. \quad [53]$$

τ_k can be calculated by integrating the above equation from $t = 0$ to $t = \tau$, which gives

$$|\varepsilon| + e^{|\varepsilon|} - 1 = \int_0^\tau q_{n-(1+k)}(t) dt \quad [54]$$

The integral $\int_0^{\tau_k} q_{n-(1+k)}(t) dt$ is determined by the dynamics in the bulk of the wave, which are governed by Eq. (50). The Laplace transform of Eq. (50) leads to a relationship similar to Eq. (37) for $m = \ell(1+k) < n$:

$$(r - q_{\ell(1+k)}(t)) = \sum_{\ell'=0}^{\ell-1} \frac{t^{\ell'} e^{-t}}{\ell'!} (r - q_{(\ell-\ell')(1+k)}(0)). \quad [55]$$

Defining $h(\ell) \equiv 1 - q_{\ell(1+k)}(0)/r$ and using the periodicity of the wave w.r.t τ_k , $q_{\ell(1+k)}((1+k)\tau_k) = q_{(\ell-1)(1+k)}(0)$, we have

$$h(\ell-1) = \sum_{\ell'=0}^{\ell-1} \frac{((1+k)\tau_k)^{\ell'} e^{-((1+k)\tau_k)}}{\ell'!} h(\ell-\ell'). \quad [56]$$

It can be checked that $h(\ell) = (1 - |\varepsilon|/r) b_k^{\ell-(n-1)/(1+k)}$ with

$$b_k = \frac{-(1+k)\tau_k}{W(-(1+k)\tau_k e^{-(1+k)\tau_k})} \quad [57]$$

satisfies the above recursion and the boundary conditions $q_{n-1}(0) = |\varepsilon|$. Defining $\beta_k \equiv 1 - 1/b_k$, we obtain

$$q_{n-1}(t) = r - (r - |\varepsilon|) e^{-\beta_k t}. \quad [58]$$

From Eq. (54), we have

$$|\varepsilon| + e^{|\varepsilon|} - 1 = r\tau_k - (r - |\varepsilon|) \frac{e^{-k\beta_k\tau_k} - e^{-(k+1)\beta_k\tau_k}}{\beta_k}, \quad [59]$$

which upon rearranging leads to

$$\tau_k = \frac{|\varepsilon| + e^{|\varepsilon|} - 1}{r} + \left(1 - \frac{|\varepsilon|}{r}\right) \frac{e^{-k\beta_k\tau_k} - e^{-(k+1)\beta_k\tau_k}}{\beta_k}. \quad [60]$$

This implicit equation can be solved numerically for τ_k . We verify the result by comparing our analytical calculation to RL simulations (Figure S4b,c). In Figure S4b, we also show the speed of the wave when the standard version of eligibility traces SARSA(λ) is used with an effective $k_{\text{eff}} = (1-\lambda)^{-1}$. Importantly, it can be shown that the second term on the right-hand-side goes to zero as $k \rightarrow \infty$. This limit corresponds to the situation in which the node n effectively receives reinforcement from future states that have already been fully reinforced to r (i.e., $\int_0^{\tau_k} q_{n-(k+1)}(t) dt = r\tau_k$). From here, we obtain the upper bound on the speed,

$$v_\infty = \tau_\infty^{-1} = \frac{r}{|\varepsilon| + e^{|\varepsilon|} - 1}. \quad [61]$$

(c) **Dyna-Q**. Dyna-Q (tabular) is a model-based RL algorithm which combines “planning” and Q -learning. It is useful to examine the learning dynamics of a model-based learning algorithm when applied to our setting. The Dyna-Q agent learns a model of the environment, $P(s', r|s, a)$, corresponding to the distribution of subsequent rewards and states for every state-action pair. At every step during navigation, in addition to updating its q -values using Eq. (63) for the current state-action pair, the agent also randomly samples n_p previously visited state-action pairs from memory, draws the subsequent state and reward from its learned model, and updates the corresponding q -values.

Dyna-Q does not significantly influence the learning dynamics in our setting due to the deterministic state transitions and rewards. Its effect is to simply scale the learning rate compared to SARSA. This scaling is because the probability, $p(s, a)$, that a particular state-action pair, (s, a) , is reinforced during each of the additional n_p planning steps is proportional to the number of times (s, a) is expected to be traversed in an episode, say, $n(s, a)$. The number of times (s, a) is reinforced in a single planning step is then $n_p p(s, a) = n_p \frac{n(s, a)}{\sum_{s', a'} n(s', a')} = n_p \frac{n(s, a)}{n_{\text{steps}}}$, where n_{steps} is the total number of state-action pairs visited in each episode. Since this planning step is applied at each step, the total *additional* number of times, $q_r(s, a)$ is updated per episode is $n_p n(s, a)$, in addition to the $n(s, a)$ regular SARSA updates. This calculation implies that (s, a) is updated by a factor $n_p + 1$ compared to the plain SARSA case, which can be interpreted as a scaling of the learning rate $\alpha \rightarrow (1 + n_p)\alpha$. Here, we have assumed that the memory of the Dyna-Q agent is not much larger than α^{-1} so that $n(s, a)$ does not change significantly over a timescale comparable to the memory size. We confirm this prediction in RL simulations of Dyna-Q agents (Figure S4a).

(d) **Q-learning**. Watkins’ Q -learning rule is closely related to SARSA with an important difference that enables off-policy learning, i.e., the agent learns the optimal q values while executing an arbitrary policy. This rule is defined by

$$\begin{aligned} q_r(s_t, a_t) &\rightarrow q_r(s_t, a_t) + \alpha \delta_t, \quad \text{where} \\ \delta_t &= r_t + \max_a q_r(s_{t+1}, a) - q_r(s_t, a_t), \text{ if } s_{t+1} \text{ is not the goal} \\ \delta_t &= r_t - q_r(s_t, a_t), \text{ if } s_{t+1} \text{ is the goal.} \end{aligned} \tag{62}$$

Note the max instead of the expectation in δ_t . We consider Q -learning using the stochastic policy considered for SARSA above. The equivalent of Eq. (25) follows by replacing $\sigma_{n-1} q_{n-1}$ in Eq. (25) with $\max(0, q_{n-1}) = q_{n-1}$ since the alternative action of turning into the side path has $q_r = 0$ and $q_{n-1} \geq 0$. We thus have

$$\begin{aligned} \dot{q}_n &= \mu_n (q_{n-1} - q_n), \\ \dot{q}'_n &= \mu'_n (q_{n-1} - q'_n). \end{aligned} \tag{63}$$

The σ_{n-1} prefactor does not play a significant role, and the results obtained for SARSA are directly applicable in both the expanding and marching regimes, which we verify through RL simulations (Figure S4b).

(e) **Q-learning (pure exploration)**. Next, we consider Q -learning with a purely explorative policy that is independent of the learned q_r values. Specifically, the policy is fixed and is given by $\ln \pi(a|s) = q_\epsilon(s, a) - \ln Z(s)$, where $q_\epsilon(s, a)$ are the intrinsic rewards as in the previous cases. This case is useful to understand the effects of the learning rule in isolation, without the effects of the feedback on behavior due to learning coming into play.

The dynamics of q_n and q'_n are given by Eq. (63). However, the factors μ_n are no longer the ones in Eq. (24). Recall that μ_n is the expected number of times per episode the agent performs the action of traversing towards the goal at node n on the direct path. This number is the expected visits to state $(n+1, n)$, $M_+(n)$ (from Eq. (21)), multiplied by the probability, $\sigma(\varepsilon)$, that the agent takes the action leading to $(n, n-1)$. We thus have $\mu_n = \sigma(\varepsilon)M_+(n)$ and, similarly, $\mu'_n = (1 - \sigma(\varepsilon))\sigma(\varepsilon')M_+(n)$. Since μ_n 's are constant over time, the differential equations Eq. (63) are linear and can be exactly solved for $q_n(t)$ in the Laplace domain. We obtain

$$q_n(t) = r \left(1 - \sum_{m=1}^n \left(\prod_{m' \neq m} \frac{\mu_{m'}}{\mu_{m'} - \mu_m} \right) \frac{e^{-\mu_m t}}{\mu_m} \right), \quad [64]$$

which unfortunately does not yield much insight. Instead, we consider the evolution of $q_n(t)$ for n large when $k_- < k_+$. From Eq. (21), we have that $\mu_n = \sigma(\varepsilon) \frac{1-k_-}{k_+ - k_-} \equiv \mu$ is a constant. The equations $\dot{q}_n = \mu(q_{n-1} - q_n)$ lead to $q_n(t) \approx rP(n, \mu t)$. This relation is approximate as μ_n is not constant for small n . The reinforcement signal thus propagates with speed

$$v = \mu = \sigma(\varepsilon) \frac{1 - k_-}{k_+ - k_-}, \quad [65]$$

which is verified in simulations (Figure S5c). This analysis further highlights that when learning is decoupled from behavior, similar to the $e^\varepsilon \gg 1$ expanding regime, the propagation of the signal is simply constrained by the local learning rule with speed proportional to the number of times the learning rule is applied at each state-action pair.

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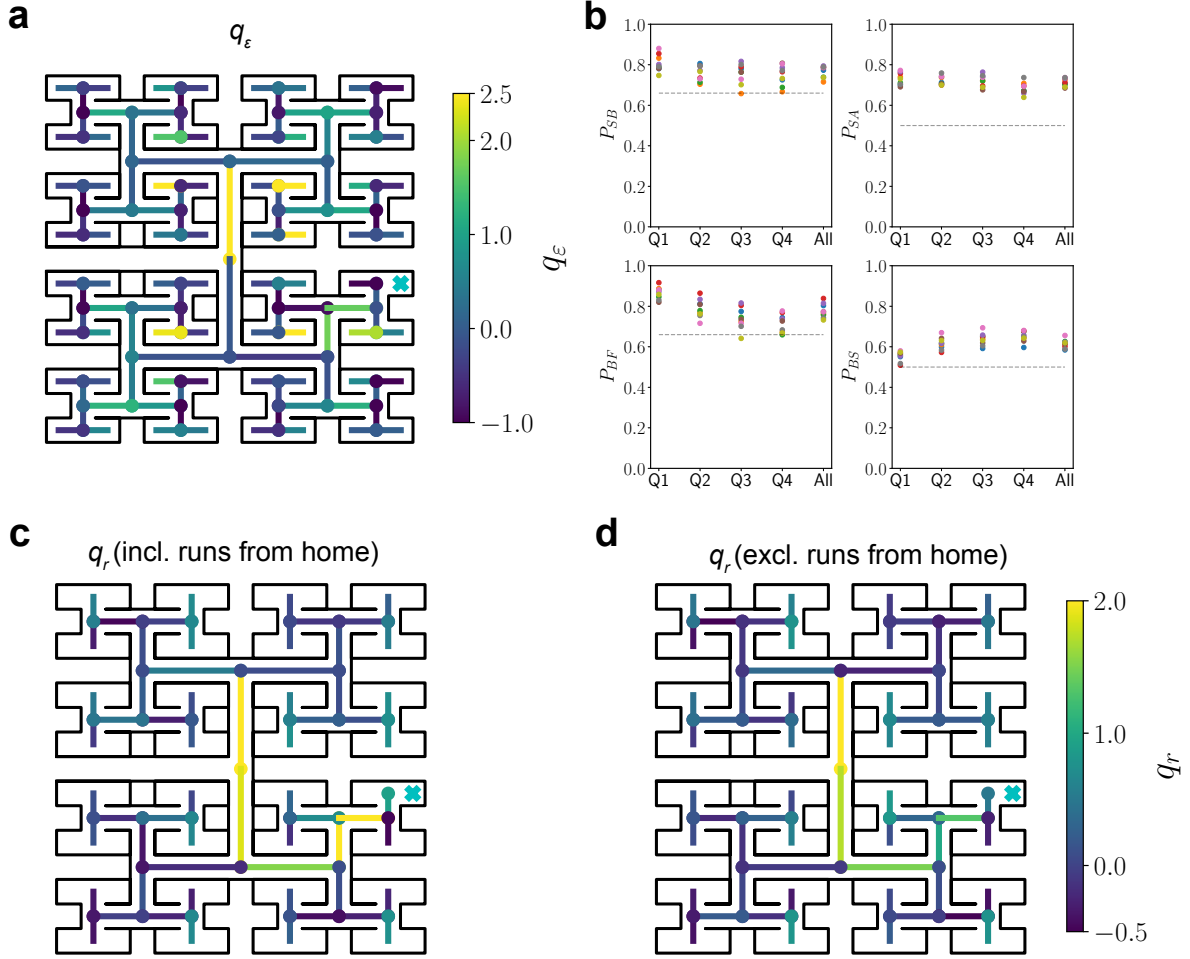


Fig. S1. (a) The q_ϵ values obtained from experimental data using MaxEnt IRL for actions leading to the reward (blue cross) relative to the other actions are shown in the top panel. For example, the color of the edge immediately after taking a right turn from home represents the q_ϵ value of taking a left turn at the next junction. (b) The exploration statistics do not change significantly over the ~ 7 hours each mouse spends in the maze. Shown here are the four exploration biases as defined by ref. (6) for the nine unrewarded animals (different colors) for the first to fourth quarters (Q1-4) and for the full duration of each experiment (All). P_{SB} : the probability that the agent goes down a branch when it comes into an intersection from the stem. P_{SA} : The probability that it alternates its turns when it comes into an intersection from the stem. P_{BF} : The probability that it does not turn back when it comes into an intersection from a branch. P_{BS} : probability that it goes up the stem when it comes into an intersection from a branch. (c,d) In the bottom two panels, we show the q_r values by comparing the behavior of rewarded and unrewarded mice. For clarity, only a subset of the q_r values for actions leading towards the reward are shown. To ensure habitual direct runs from home to goal do not significantly bias the estimated q_r values close to the reward, we show the estimates from data when runs from home are included (left) and excluded (right).

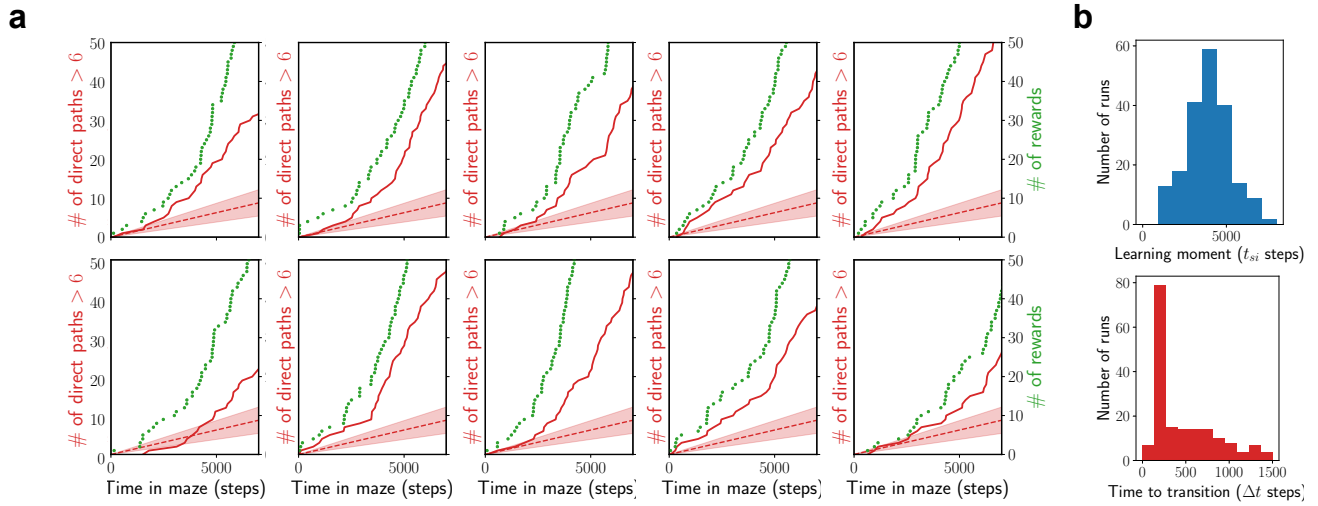


Fig. S2. (a) Learning curves from ten random runs of the RL simulations. The dashed lines and solid fill correspond to the average $\pm$ one standard deviation in the rate of direct paths for unrewarded agents. (b) To quantify the variability across realizations, we repeat the RL simulations for 200 runs and fit the rate of the direct path for each run to a logistic function, $r(t) = r_i + \Delta r \sigma\left(\frac{t - t_{si}}{\Delta t}\right)$ using the maximum likelihood method described in ref. (6). The histograms for the best fit t_{si} and Δt are shown. Note the sharp peak at $\Delta t \approx 150$ steps (≈ 3 episodes). The peak occurs when the reinforcement wave reaches all intersections of distance 6 from the reward at the same time.

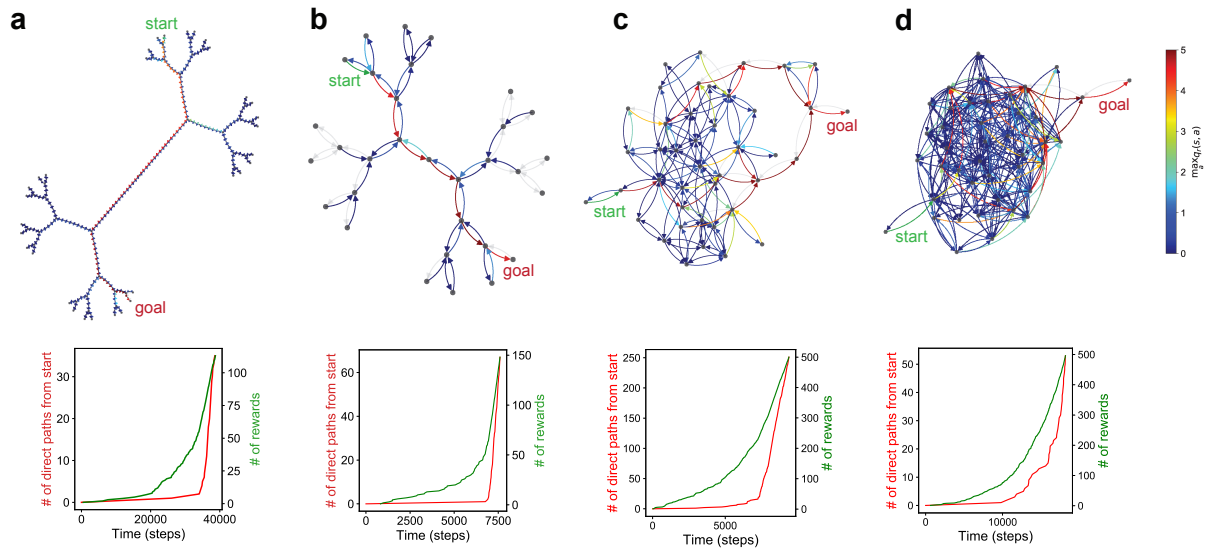


Fig. S3. Learning curves for different graph architectures. In all simulations, we use a standard softmax policy ($\beta = 1$) with $\alpha = 0.1$, $r = 5$, $\gamma = 1$. The colors show the q_r value of the best action at each state (directed edge). In panels b,c,d, we use SARSA(0) and in panel a we use SARSA(λ) where the eligibility trace parameter is $\lambda = 0.95$. The gray edges have maximum q_r value less than 10^{-3} . (a) A depth-5 binary tree where locations within a corridor are included in the state space, reflecting the maze experiments of ref. (6). The lowest level of the tree contains one additional state within the corridor. Other levels have twice the number of states as the level beneath them. See also Movie S3. (b) A depth-4 binary tree where the agent is allowed to reverse its direction. See also Movie S4. (c,d) A sparse and dense random graph (Erdős-Renyi) with 42 nodes and $p = 0.08, 0.2$ respectively. The start and goal are attached to the two nodes that are farthest apart (8 and 6 nodes apart respectively).

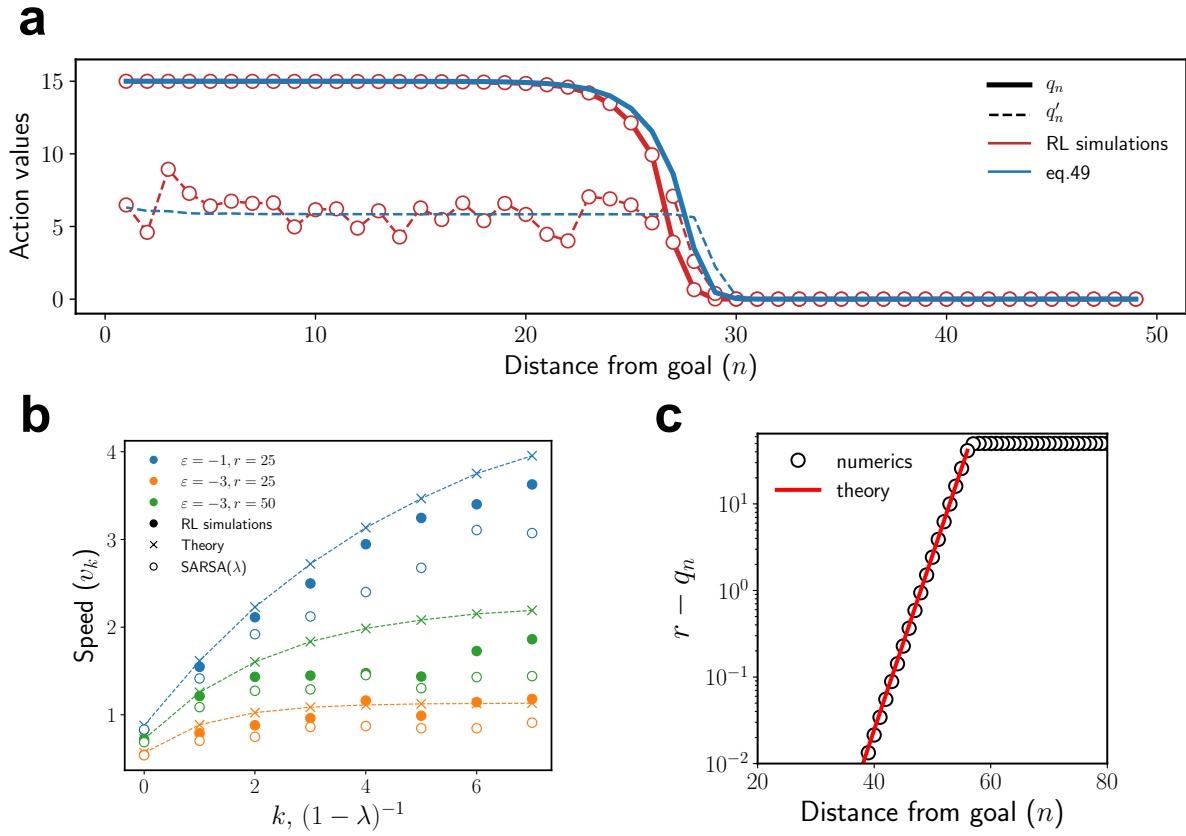


Fig. S4. Numerics and theory for RL with eligibility traces. (a) The q_n, q'_n from RL simulations align closely with numerics Eq. (49). (b) The speed from RL simulations and theory for a range of values of k, ε, r . Shown also is the speed of the wave when the standard version of eligibility traces SARSA(λ) is used with an effective $k_{\text{eff}} = (1 - \lambda)^{-1}$. (c) The shape of the wave front from numerics matches the theory prediction Eq. (58). Here $k = 2, \varepsilon = -2, r = 15$.

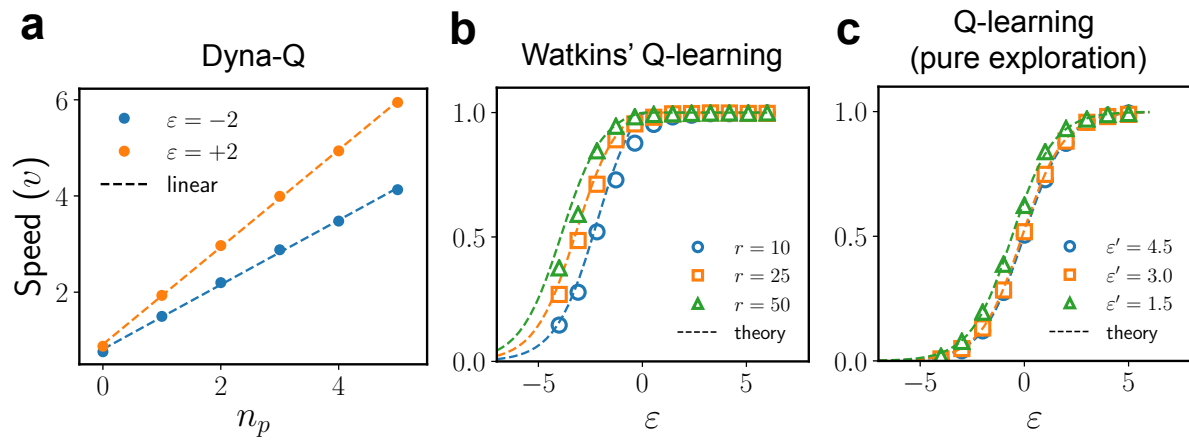


Fig. S5. Theoretical predictions of the speed for various RL rules compared to those obtained from simulations.